Timbol Steven B. ICT-106

Problem Set 2: Slope and equation of the line.

1. Find the slope of the line passing through the points $A(\overset{x_1}{9},\overset{y_1}{1})$ and $B(\overset{x_2}{4},\overset{y_2}{3})$

$m = \frac{y_2 - y_1}{x_2 - x_1}$ $m = \frac{3-1}{4-9}$ $m = \frac{2}{-5}$ $=$ $\boxed{m = -\frac{2}{5}}$

2. What is the equation of the line in general form that passes through the point $(2, \overset{y_1}{-3})$ with the slope of 5.

$y - y_1 = m(x - x_1)$ $y - (-3) = 5(x - 2)$ $y + 3 = 5x - 10$

$\boxed{5x - y - 13 = 0}$

3. Find the equation of the line in general form whose y-intercept is $-7$ and the slope is $-\frac{2}{3}$

$y = mx + b$ $y = -\frac{2}{3}x - 7$ $3y = -2x - 21$ $\boxed{2x + 3y + 21 = 0}$

4. What is the equation of the line in general form whose x-intercept is $-2$ and y-intercept is 4?

$\frac{x}{a} + \frac{y}{b} = 1$
$(-2)$ $(4)$

$\frac{x}{-2} + \frac{y}{4} = 1$ $-2y + 4x = -8$ $\boxed{4y - 2x + 8 = 0}$

5. Find the equation of the line that passes through the points $(\overset{x_1}{-1}, \overset{y_1}{2})$ and $(\overset{x_2}{-3}, \overset{y_2}{-1})$.

$m = \frac{-1-2}{-3-(-1)}$ $m = \frac{-3}{-2} = \frac{3}{2}$

$m = \frac{y_2 - y_1}{x_2 - x_1}$

Point shape form: $y - y_1 = m(x - x_1)$ $y - 2 = \frac{3}{2}(x - (-1))$ $2y - 4 = 3x + 3$

$\boxed{= 3x - 2y + 7 = 0}$

6. What is the equation of the vertical line passing through the point $(-8, 0)$?

$\boxed{x = -8}$